MATH 2228-601, SUMMER II 2025, MOCK MIDTERM 2

INSTRUCTIONS

1. Do not open the exam paper or start working until you are told to do so.

2. Bubble in your name as DOE John S (if it is John S. Doe), bubble in your student ID number.

3. NO NOTES OR BOOKS ARE ALLOWED. You may not communicate with others during the exam. Please be considerate and refrain from making noise. Switch off any beepers, cell phones, etc.

4. Bubble in your answer to, say, Question 4, on line 4.

5. The exam is self-explanatory. The instructor will not interpret the questions for you. You may, however, ask for another copy of the exam if your copy is illegible.

6. Have your calculator in **RADIAN MODE**. You can set radian mode by pressing a button called MODE, SET UP, or DRG (degree, radian, gradian).

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1. Let Q_1 be the minimum, Q_2 the first quartile, Q_3 the median, Q_4 the third quartile, and Q_5 the maximum of the list below.

152, 689, 608, 717, 688, 857, 469, 318, 127, 559, 610, 661, 850, 633, 322, 469, 391, 447, 559, 828, 782, 160, 424

Let $Q = \ln(3+|Q_1|+2|Q_2|+3|Q_3|+4|Q_4|+5|Q_5|)$. Then $T = 5\sin^2(100Q)$ satisfies:— (A) $0 \le T < 1$. — (B) $1 \le T < 2$. — (C) $2 \le T < 3$. — (D) $3 \le T < 4$. — (E) $4 \le T \le 5$.

2. Let Q_1 be the mean of x, Q_2 the sample standard deviation of x, Q_3 the mean of y, Q_4 the sample standard deviation of y, and Q_5 the linear correlation coefficient of x and y, where the sequences x and y are as follows:

x: $-48, -52, -89, 35, -18, -62, -96, -47, 84, -93, 98, 38,$

y: $-82, -7, -63, 14, -57, 91, -15, 36, 6, -77, 79, -52.$

Let $Q = \ln(3+|Q_1|+2|Q_2|+3|Q_3|+4|Q_4|+5|Q_5|)$. Then $T = 5\sin^2(100Q)$ satisfies:— (A) $0 \le T < 1$. — (B) $1 \le T < 2$. — (C) $2 \le T < 3$. — (D) $3 \le T < 4$. — (E) $4 \le T \le 5$.

3. Let Q_1 be the slope and Q_2 the intercept of the linear regression line $y = ax + b$, where the sequences x and y are as follows:

x: $80, -33, -24, 64, -18, 76, -43, -99, 12, 42, -12, 0,$

y: $10, -47, 89, -22, 45, 35, 93, -100, 57, -27, -7, -95$.

Let $Q = \ln(3 + |Q_1| + 2|Q_2|)$. Then $T = 5\sin^2(100Q)$ satisfies:— (A) $0 \le T < 1$. — (B) $1 \le T < 2$. — (C) $2 \le T < 3$. — (D) $3 \le T < 4$. — (E) $4 \le T \le 5$.

4. Let Q_1 be the slope, Q_2 the intercept of the linear regression line $y = ax + b$, and Q_3 the prediction $\hat{y}_0 = ax_0 + b$ for $x_0 = 20.77$, where the sequences x and y are as follows:

x: $94, -83, 15, -85, 22, 82, 10, -19, 21, -57, 57, 92$,

y: $52, 45, -7, 84, -34, -49, -82, -42, 95, 17, -84, -54$.

Let $Q = \ln(3 + |Q_1| + 2|Q_2| + 3|Q_3|)$. Then $T = 5\sin^2(100Q)$ satisfies:— (A) $0 \le T < 1$. — (B) $1 \le T < 2$. — (C) $2 \le T < 3$. — (D) $3 \le T < 4$. — (E) $4 \le T \le 5$.

5. Let Q_1 be the slope, Q_2 the intercept of the linear regression line $y = ax + b$, and Q_3 the coefficient of determination, where the sequences x and y are as follows:

x: $51, 51, 28, -16, 52, 82, -43, 14, -7, 8, 68, 5$,

y: $-80, -78, -49, 33, 92, -25, 51, 58, 88, -6, 73, 37$.

Let $Q = \ln(3 + |Q_1| + 2|Q_2| + 3|Q_3|)$. Then $T = 5\sin^2(100Q)$ satisfies:— (A) $0 \le T < 1$. — (B) $1 \le T < 2$. — (C) $2 \le T < 3$. — (D) $3 \le T < 4$. — (E) $4 \le T \le 5$.

6. The table below shows how many parts in a lot fail or do not fail when made of one of three metal alloys. Relative to the table below, let Q_1 be the probability that a part fails given that it is made of metal A, Q_2 the probability that a part fails, Q_3 the probability that a part does not fail given that it is not made of metal B, and Q_4 the probability that a part fails or it is made of metal C.

	metal A	metal B	metal C
part fails	182	230	168
part does not fail	505	783	352

Let $Q = \ln(3 + |Q_1| + 2|Q_2| + 3|Q_3| + 4|Q_4|)$. Then $T = 5\sin^2(100Q)$ satisfies:— (A) $0 \le T < 1$. — (B) $1 \le T < 2$. — (C) $2 \le T < 3$. — (D) $3 \le T < 4$. — (E) $4 \le T \le 5$.

7. Suppose that the events E, F, G are independent and have probabilities $P(E) = 0.258$, $P(F) = 0.292$, $P(G) = 0.11$. Denote the complement of an event H by H^c. Let $Q_1 = P(E \cap F^c \cap G)$, $Q_2 = P(E^c \cap G)$, and $Q_3 = P(E \cap F \cap G^c)$. Let $Q = \ln(3 + |Q_1| + 2|Q_2| + 3|Q_3|)$. Then $T = 5\sin^2(100Q)$ satisfies:— (A) $0 \le T < 1$. — (B) $1 \le T < 2$. — (C) $2 \le T < 3$. — (D) $3 \le T < 4$. — (E) $4 \le T \le 5$.

8. Let a letter be one of the 26 letters A to Z. Let a digit be one of the 10 digits 0 to 9. Let Q_1 be the count of all possible sequences, with repetitions allowed, of 7 letters and 20 digits. Let $Q = \ln(3 + |Q_1|)$. Then $T = 5\sin^2(100Q)$ satisfies:— (A) $0 \le T < 1$. — (B) $1 \le T < 2$. — (C) $2 \le T < 3$. — (D) $3 \le T < 4$. — (E) $4 \le T \le 5$.

9. Let Q_1 be the number of combinations of length 10 of 36 distinct elements. Let

$Q = \ln(3 + |Q_1|)$. Then $T = 5\sin^2(100Q)$ satisfies:— (A) $0 \le T < 1$. — (B) $1 \le T < 2$. — (C) $2 \le T < 3$. — (D) $3 \le T < 4$. — (E) $4 \le T \le 5$.

10. Let Q_1 be the number of all the ways in which we can select a committee consisting of a chairman, a vice chairman, a treasurer, and 10 ordinary members (whose order does not matter) out of 25 people. Let $Q = \ln(3 + |Q_1|)$. Then $T = 5\sin^2(100Q)$ satisfies:— (A) $0 \le T < 1$. — (B) $1 \le T < 2$. — (C) $2 \le T < 3$. — (D) $3 \le T < 4$. — (E) $4 \le T \le 5$.

11. Let X be the number of successful rocket launches in 636 independent trials with success probability 0.927. Using the binomial distribution, let Q_1 the probability that $X \ge 587$, Q_2 the probability that $X \le 596$, and Q_3 the probability that $584 \le X \le 599$. Let $Q = \ln(3 + |Q_1| + 2|Q_2| + 3|Q_3|)$. Then $T = 5\sin^2(100Q)$ satisfies:— (A) $0 \le T < 1$. — (B) $1 \le T < 2$. — (C) $2 \le T < 3$. — (D) $3 \le T < 4$. — (E) $4 \le T \le 5$.

12. Let X be a normal random variable with mean 258 and standard deviation 14.82. Let Q_1 the probability that $X \ge 241$, Q_2 the probability that $X \le 291$, and Q_3 the probability that $229 \le X \le 267$. Let $Q = \ln(3 + |Q_1| + 2|Q_2| + 3|Q_3|)$. Then $T = 5\sin^2(100Q)$ satisfies:— (A) $0 \le T < 1$. — (B) $1 \le T < 2$. — (C) $2 \le T < 3$. — (D) $3 \le T < 4$. — (E) $4 \le T \le 5$.